MTH 201 Chapter 3: Infinite Series

3.1.1 Sequences Worksheet

Determine whether each of the following sequences converges or diverges.

1. $\left\{\dfrac{2n}{3n-1}\right\}_{n=1}^{\infty}$

   Let $f(x)=\dfrac{2x}{3x-1}$. We see that $\lim\limits_{x\to\infty}\dfrac{2x}{3x-1}\overset{H}{=}\lim\limits_{x\to\infty}\dfrac{2}{3}=\dfrac{2}{3}$. Since $\lim\limits_{x\to\infty}f(x)=\dfrac{2}{3}$, then $\lim\limits_{n\to\infty}\dfrac{2n}{3n-1}=\dfrac{2}{3}$.

2. $\{6-3n\}_{n=1}^{\infty}$

   Let $f(x)=6-3n$. We see that $\lim\limits_{x\to\infty}6-3n=-\infty$. Since $\lim\limits_{x\to\infty}f(x)=-\infty$, then $\lim\limits_{n\to\infty}6-3n=-\infty$.
   Therefore, the given sequence diverges.

3. $\left\{\dfrac{n}{e^n}\right\}_{n=1}^{\infty}$

   Let $f(x)=\dfrac{x}{e^x}$. We see that $\lim\limits_{x\to\infty}\dfrac{x}{e^x}\overset{H}{=}\lim\limits_{x\to\infty}\dfrac{1}{e^x}=0$. Since $\lim\limits_{x\to\infty}f(x)=0$, then $\lim\limits_{n\to\infty}\dfrac{n}{e^n}=0$.

4. $\left\{\sqrt[n]{n}\right\}_{n=1}^{\infty}$

   Let $f(x)=\sqrt[x]{x}=x^{1/x}$. We see that $\lim\limits_{x\to\infty}x^{1/x}=\lim\limits_{x\to\infty}e^{\ln\left(x^{1/x}\right)}=e^{\lim\limits_{x\to\infty}\frac{1}{x}\ln(x)}=e^{\lim\limits_{x\to\infty}\frac{\ln(x)}{x}}\overset{H}{=}e^{\lim\limits_{x\to\infty}\frac{1}{x}}=e^0=1$.
   Since $\lim\limits_{x\to\infty}f(x)=1$, then $\lim\limits_{n\to\infty}\sqrt[n]{n}=1$.

5. $\left\{(-1)^n \frac{1}{n}\right\}_{n=1}^{\infty}$

Let $f(x) = \left|(-1)^x \frac{1}{x}\right| = \frac{1}{x}$. We see that $\lim_{x\to\infty} \frac{1}{x} = 0$. Since $\lim_{x\to\infty} f(x) = 0$, then $\lim_{n\to\infty} \left|(-1)^n \frac{1}{n}\right| = 0$. Since the absolute value of the sequence converges to 0 we know that $\lim_{n\to\infty} (-1)^n \frac{1}{n} = 0$.

6. $\left\{(-1)^n \frac{4n}{1-7n}\right\}_{n=1}^{\infty}$

The positive terms of this sequence converge to $\lim_{n\to\infty} \frac{-4n}{1-7n} = \frac{4}{7}$. The negative terms of this sequence converge to $\lim_{n\to\infty} \frac{4n}{1-7n} = -\frac{4}{7}$. Since the negative and positive terms of the sequence converge to different values the sequence must diverge.

7. $\left\{(-1)^n \frac{1}{4^n}\right\}_{n=1}^{\infty}$

Let $f(x) = \left|(-1)^x \frac{1}{4^x}\right| = \frac{1}{4^x}$. We see that $\lim_{x\to\infty} \frac{1}{4^x} = 0$. Since $\lim_{x\to\infty} f(x) = 0$, then $\lim_{n\to\infty} \left|(-1)^n \frac{1}{4^n}\right| = 0$.

Since the absolute value of the sequence converges to 0 we know that $\lim_{n\to\infty} (-1)^n \frac{1}{4^n} = 0$.